

Title

Phase 2 randomized trial protocol: vitamin D metabolite supplementation to mitigate kidney injury in intensive care patients

Abstract

Background: Acute kidney injury in critically ill populations carries substantial morbidity and mortality. Preclinical studies suggest vitamin D metabolites possess renoprotective and immunomodulatory properties. This protocol describes a phase 2 randomized investigation of two activated vitamin D formulations in high-risk intensive care patients. **Methods:** One hundred fifty adults with critical illness and baseline kidney dysfunction risk factors will be randomized to one of three arms: calcifediol (25-hydroxyvitamin D derivative), calcitriol (1,25-dihydroxyvitamin D derivative), or placebo, administered orally. Primary composite outcome combines 7-day mortality, need for renal replacement therapy, and magnitude of serum creatinine elevation using a pre-specified ranking procedure. Secondary outcomes include incidence of new kidney injury, laboratory biomarkers, and transcriptomic analysis of immune cells. Safety monitoring focuses on hypercalcemia incidence. Study design incorporates intent-to-treat analysis with pre-specified subgroup interrogation. Findings will inform subsequent phase 3 trial design and optimal vitamin D formulation selection.